

Shri G. S. Institute of Technology and Science, Indore
Department of Applied Mathematics and Computational Science
B.Tech. I Year (Group A: BM, CE, CS, EC, EE)(July-December 2025)
Mid Term Test-II MA10021: Mathematics for Engineers

Time: 1 hour

Max. Marks: 20

Note: Attempt all questions. All questions carry equal marks.

S.No	Questions	Marks	CO	BL
Q.1	Evaluate $\int_0^{\infty} \sqrt{x} e^{-3\sqrt{x}} dx$.	04	CO2	2
Q.2	Evaluate $\int \int_R y dx dy$, where R is the region bounded by the parabolas $y^2 = 4x$ and $x^2 = 4y$.	04	CO2	2
Q.3	For what values of λ and μ do the system of equations $x + y + z = 6, x + 2y + 3z = 10, x + 2y + \lambda z = \mu$ have (i) no solution (ii) a unique solution (iii) an infinite number of solutions.	04	CO3	2
Q.4	a) Find the eigen values of the matrix $A = \begin{bmatrix} 1 & -2 \\ -5 & 4 \end{bmatrix}$	02	CO3	2
	b) Find the rank of the matrix $\begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$	02		
Q.5	a) Define fuzzy set. b) If fuzzy set $A = \{(x_1, 0.9), (x_2, 0.8), (x_3, 0.7), (x_4, 0.2)\}$ and $B = \{(x_1, 0.5), (x_2, 0.4), (x_3, 0.6), (x_4, 0.2)\}$ then find out $A \cup B, A \cap B$ and complement of A.	04	CO5	1,2

